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MyDesign Portfolio

Element B Initial Template

How to use this template

Students: The template below is one way to structure your research findings around past and current attempts to solve the problem, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own research into prior design attempts, be sure to delete the instructions.

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Filament Rack>

Researching in the Class

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Source - Mr. L

The rack contains several horizontal pegs that store a couple rolls per peg. It is made up of pieces of quarter inch plywood and .25-.5 inch diameter dowels. This rack could hold the potential for a sorting system, but had no apparent sorting. Different colors and sizes were all mixed up together with no organization.

Analysis

This compact storage system was inefficient and underdeveloped. There was a complete lack of organization and sorting. All the filaments were very visible so this allowed for students and teachers to observe their many options or filament colors and types. This storage was accessible to all students and allowed easy access to a variety of filaments.

Prior Design Attempt #2 - <Designated Storage Shelf>

Researching in the Class

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).

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-Carter Stone 2024

Details

All the filaments regardless of size or color are stacked haphazardly under the 3d printers. The filaments are taking up a lot of space and that space is needed for more 3d printers. Maximum 200 rolls of filament is hard to sort through, causing people to ask Mr. L to find the filament for them. The shelf is roughly 4.5 x 2 x 2.5 feet in size.

Analysis

This design was a bare minimum for the filament so that all the filament was in one place and it succeeded in that but did not work with anything else. Needed filament was hard for

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students to find and resulted in Mr. L to have to sort through the stacks of filament to find what the students needed. The storage needs to be accessible by students and to make locating the needed filament to be easy.

Prior Design Attempt #3 - <Close-to-Site Upright Storage>

Source

https://adms.fnal.gov/3D_Printing.htm

Image



Details

At the Mechanical Support department in the Fermi National Laboratory, they prioritize rapid prototyping and quick iterative design and development. They keep 3d printer spools



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close to the printer, as a part of a mobile station containing everything needed for prototyping. The spools are stored standing upright at the bottom of the mobile shelf, supported by the lip of the shelf.

Analysis

While the design is successful at supporting a self contained unit with a 3D printer, it doesn't work as well for other purposes. If the design were scaled up to more spools it would be difficult to reach out and grab them, running into similar issues as just having a disorganized system. Also, the design still takes up a lot of horizontal space, which will need to be used by the new 3D printers in Mr. L's workshop. It is likely that the storage for the spools will need to be separated from the 3D printers, which would defeat many of the advantages of this design.

